1. Background/Theory

SAR

Synthetic Aperture Radar (SAR) is an active microwave sensor, meaning that it emits energy of its own and captures the energy reflected off the surface (backscatter). SAR does not require sunlight and microwaves penetrate cloud, allowing for the imaging of the ground in situations where optical sensors cannot. The backscatter also differs depending on the surface roughness, moisture and vegetation structure, and thus contains information which is not contained in optical reflectance.

Sentinel-1 supports dual Polarisation data in Interferometric Wide (IW) mode and works in C-band. VV (vertical transmit, vertical receive) is sensitive to surface scattering and soil scattering while VH (vertical transmit, horizontal receive) is sensitive to volume scattering from vegetation canopies. A Sentinel-1 Ground Range Detected (GRD) product is a calibrated and dB expressed product derived in Google Earth Engine (GEE). SAR is likely to enhance crop analysis when used in conjunction with optical information like NDVI in the presence of cloud cover.

Optical Features and NDVI

Optical satellite data gives information about the Earth's surface by sensing reflected solar radiation in specific wavelengths. For our project, we applied the existing optical data for Argentina and Brazil for researching the vegetation parameters and comparing corn, soybean, and wheat crops.

Sentinel-2 provides a variety of spectral bands, which reflect different features of vegetation and surface. The red band (B04) and the near-infrared band (B08) were used to calculate NDVI. Generally, vegetation absorbs red radiation and reflects near-infrared one, thus NDVI can be applied for studying vegetation greenness .

The formula for calculating NDVI is:

NDVI = B08-B04/B08+B04

where B08 stands for near-infrared band and B04 for red one. In our project, NDVI was calculated to explore the NDVI distribution for the existing optical samples and to compare three categories of crops.

2. Materials and Tools

Materials and Tools (SAR Extraction)
Datasets:
Google Earth Engine (COPERNICUS/S1_GRD) Sentinel-1 SAR GRD (IW mode, VV and VH polarisations, in dB). For each sampled pixel, used for the extraction of backscatter at 10 m.
Images from Sentinel-2 from COPERNICUS/S2_SR_HARMONIZED and COPERNICUS/S2_HARMONIZED from Google Earth Engine. For validating the field locations from the band values of the data set.
The data provided for Brazil and Argentina comes from YieldSAT field-level data (https://yieldsat.github.io/) in the form of two CSVs (Brazil_with_coordinates.csv, Argentina_with_coordinates.csv). They hold 100 pixel samples (50 per country) from 6 fields, with crop labels, optical band values, dates, pixel indices (row/col) and encoded coordinates (coord_x/y/z).

Tools
Google Earth Engine Code Editor (JavaScript): Making point geometries, filtering Sentinel-1 images, sampling VV and VH, visualising the two bands separately, and exporting the results.
The imported tables for Brazil and Argentina are stored as Earth Engine assets.
Python (pandas, NumPy): looking at the CSVs, decoding the coordinate vectors, and checking the exported table.
Google Drive: getting the exported CSV file.

Component	Details
Input data	Argentina.csv and Brazil.csv
Programming language	Python
Environment	Google Colab / Jupyter Notebook
Libraries	Pandas, NumPy, Matplotlib
Output file	Margi_Week3_Optical_Features.csv

3. **Methodology**

SAR

Methodology

1.) Location recovery: The provided CSVs were not provided with longitude/latitude, so during import GEE created empty geometries for all 100 records. The coord_x, coord_y and coord_z were determined to be unit-length vectors (all with norm 1.0 for all records), which is consistent with the following equations: $x = \cos(\varphi)\cos(\lambda)$, $y = \cos(\varphi)\sin(\lambda)$, $z = \sin(\varphi)$, where φ and λ are in degrees used as radians. This encoding is derived from the data, but not described in the dataset source.

2.)Validation: Decoding leaves repeated candidate locations every 6.28° of longitude, therefore, each field's location was chosen by comparing the Sentinel-2 values at the decoded locations with the values in the dataset. In Brazil field 10, there were 4 bands (B2, B3, B4, B8) that were exact matches. There was a strong correlation between the 16-pixel NIR pattern and Argentina field 468 (0.93), and Argentina field 52 provided close band values.

3.)Geometry: Point geometries were created for the 50 pixel samples of the three verified fields (17 Brazil, 33 Argentina), and all are

non-empty. The 100 records are actually pixel samples (row/col) of 6 fields, not 100 separate fields.
4.)Sentinel-1 filtering: COPERNICUS/S1_GRD images were filtered by location, IW mode and VV and VH. The one image in each field most like the date of the dataset (within ±12 days) was chosen; this ensures a consistent direction of orbit within each field.
5.)Extraction: Sampling was performed at 10 m scale, at every pixel, of VV and VH in dB. If the value of the variable was missing, it was retained as missing and identified (valid_VV, valid_VH, status). Zero was never used to fill in missing values. These scenes are in UTM projection (latest code: EPSG:32722 for Brazil; EPSG:32720 for Argentina), and have a nominal pixel size of 10 m.

Optical Feature Extraction

The first step was loading the Argentina and Brazil CSV datasets in Python. The data records, crops, missing values, and duplicates were observed for an analysis of data quality.

Next, after observing the data quality, we computed NDVI using red and near-infrared bands. For each record, the NDVI calculation formula was used to calculate NDVI value, which was added as a new column in the dataset.
The workflow follows:

- Argentina and Brazil datasets
- NDVI Calculation
- Processed features
- Generated visualisations

The processed data was saved in the csv. Next, we created three plots - NDVI histogram for looking at the distribution of NDVI value, box plot of crops for analyzing three types of crops, and box plot of optical bands for looking at selected spectral bands.
The dataset alignment and validation phase was carried out to evaluate the representativeness and consistency of the selected optical datasets: images from the YieldSAT, in terms of the precision and saturation with which they could be used in the workflow of the crop-classification. The csv files were analyzed by using Python and Panda.
The entire validation process involves:

- Review of Recorded Numbers

- Unique Field Identifiers
- Crop Class Distribution
- Coordinates check
- Observation dates

4. Results - Optical Data Analysis

The results were summarized into a preliminary validation report and downloaded as a csv file. In the preliminary validation, 50 records were investigated from Argentina and 50 records were investigated from Brazil, resulting in a total of 100 records. There were 3 different field identifier samples in each country. No missing field identifiers, invalid observation dates or missing coordinate values were found for the samples examined.

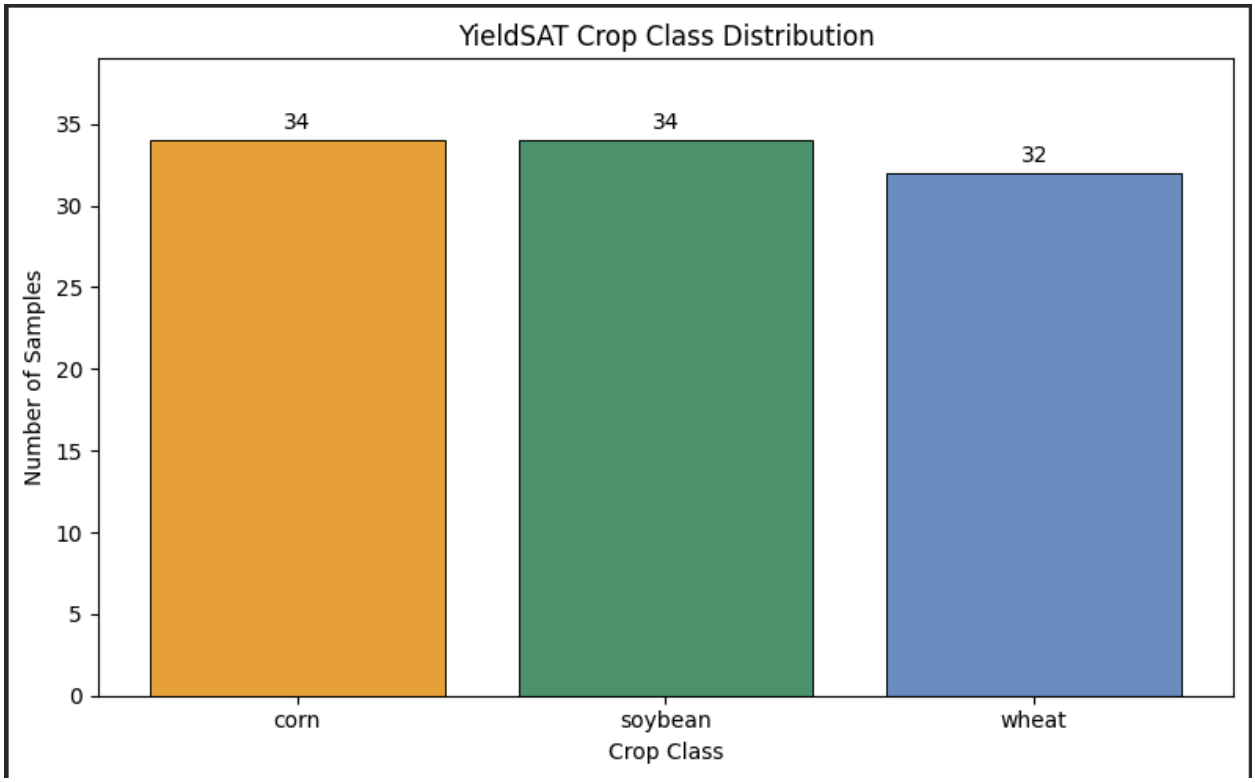
The sampling dates for the Argentina sample were between 19 September 2016 and 28 October 2021, while the sampling dates for the Brazil sample were from 16 March 2017 until 26 October 2022.

In total 34 corn records, 34 records for soybeans, and 32 wheat records were seen among the crop class distribution. This distribution suggests the selected sample size was normal and included all 3 crop classes with relatively similar sample sizes.

The distribution of the three classes of crops in the 100-record sample is illustrated in the below figure.

NDVI Statistics:

Metric	**Value**
Minimum	0.188
Maximum	0.676
Mean	0.357
Standard Deviation	0.148



The results collaborate the completeness of the metadata fields inspected. However, the checks are indicative of the data validation prior to validating optical/SAR successful alignment. For correlation Sentinel-1 should be further validated with reference to optical observations acquired at comparable dates, places and spatial resolutions.

5. Discussions:

Our first analysis showed changes in NDVI values and clear differences between the crop types. This gave us a sense of what the optical data can tell us. Still we have to check on the accuracy of crop classification.

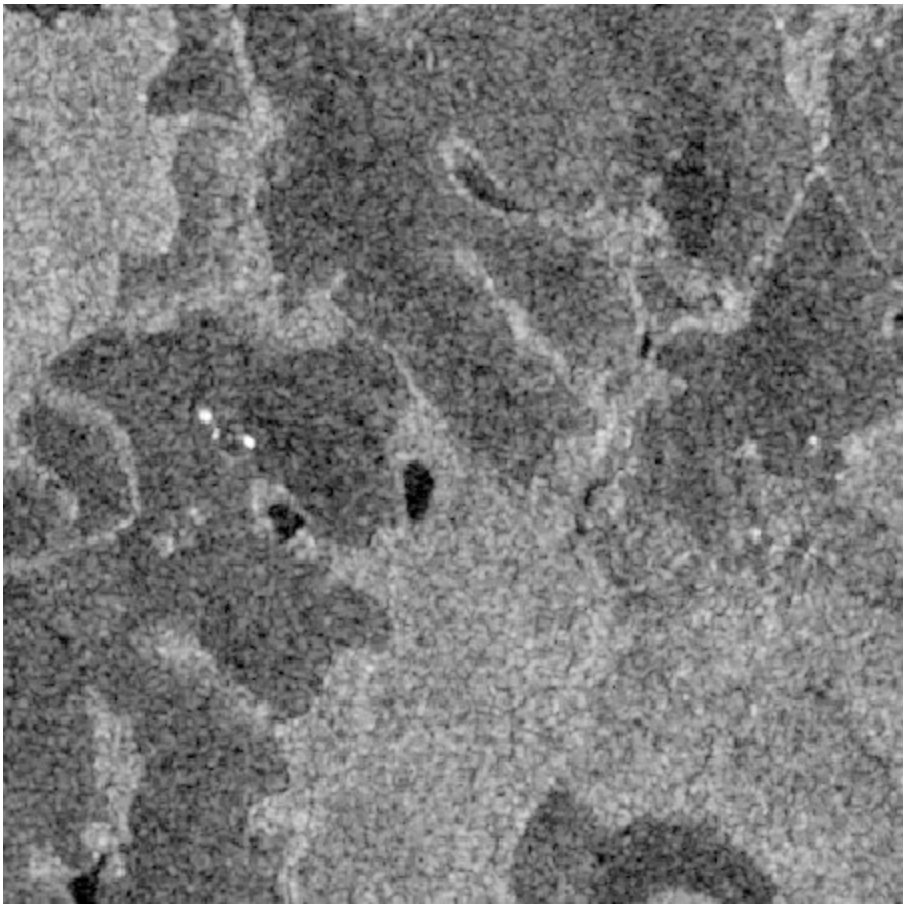
Updated SAR Results for the Team

We got VV and VH values for 50 pixel samples from Argentina and Brazil. The average VV value came out to about −10.123 dB. The average VH value was −16.813 dB. We still need to confirm that these SAR samples line up with the optical dataset before we proceed with feature fusion.

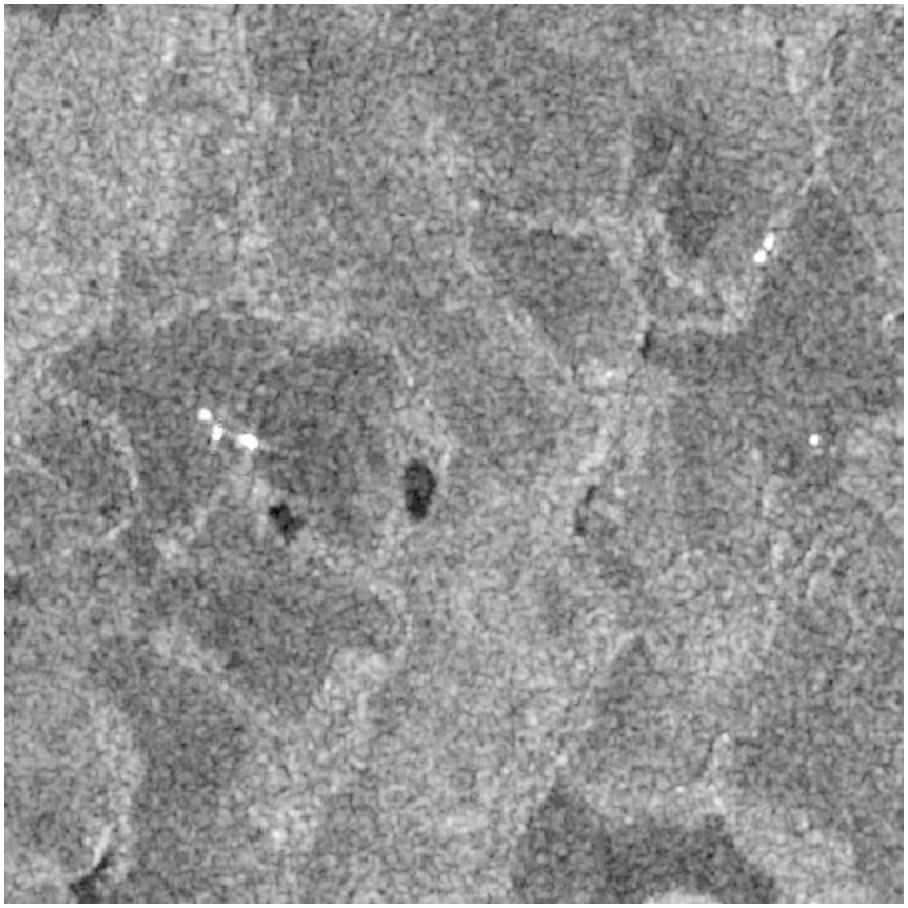
Results

The dataset has a total of 100 pixel samples, with 50 being found and retrieved. These are from three fields: Brazil field 10, Argentina fields 52 and 468. No missing, masked or zero values were returned for all 50 samples for VV and VH values from Sentinel-1. Here, the image from each field was taken with a single descending orbit, within the 12-day window from the date on the dataset.The best available Sentinel-1 image for Brazil field 10 (dataset date 26 October 2022) was from 20 October 2022, 5.6 days prior. Across its 17 pixels, VV averaged −8.88 dB (range −10.46 to −5.81) and VH averaged −15.23 dB (range −18.11 to −11.98). The image shown is from Argentina field 52 (dataset date 28 October 2021), 2.6 days earlier on relative orbit 39. Its 17 pixels averaged −12.51 dB in VV (range −14.55 to −10.05) and −21.85 dB in VH (range −28.37 to −17.03). In both polarisations, this was the least backscattering field. The image was taken for Argentina field 468 (dataset date 19 April 2021) on 16 April 2021, 2.6 days earlier and on relative orbit 39. Its 16 pixels averaged −8.90 dB in VV (range −10.23 to −7.10) and −13.14 dB in VH (range −16.16 to −10.48). In all 50 samples, VH was found to be lower than VV in all cases with means of −10.12 dB for VV and −16.81 dB for VH. The nominal pixel size of the Sentinel-1 scenes is 10m and they are in UTM projection (EPSG:32722 for Brazil and EPSG:32720 for Argentina).The Sentinel-2 were used to verify the locations that were decoded. At the decoded pixel, four bands (B2, B3, B4 and B8) compared to the dataset exactly matched. The near-infrared (0.93 across 16-pixels, 0.88 for the red band) matched the dataset in Argentina field 468. In Argentina field 52 the band values were near the dataset values, as in the near-infrared band, which had values of 3068 and 3062, respectively.

Brazil field 10 (2022-10-20, descending):

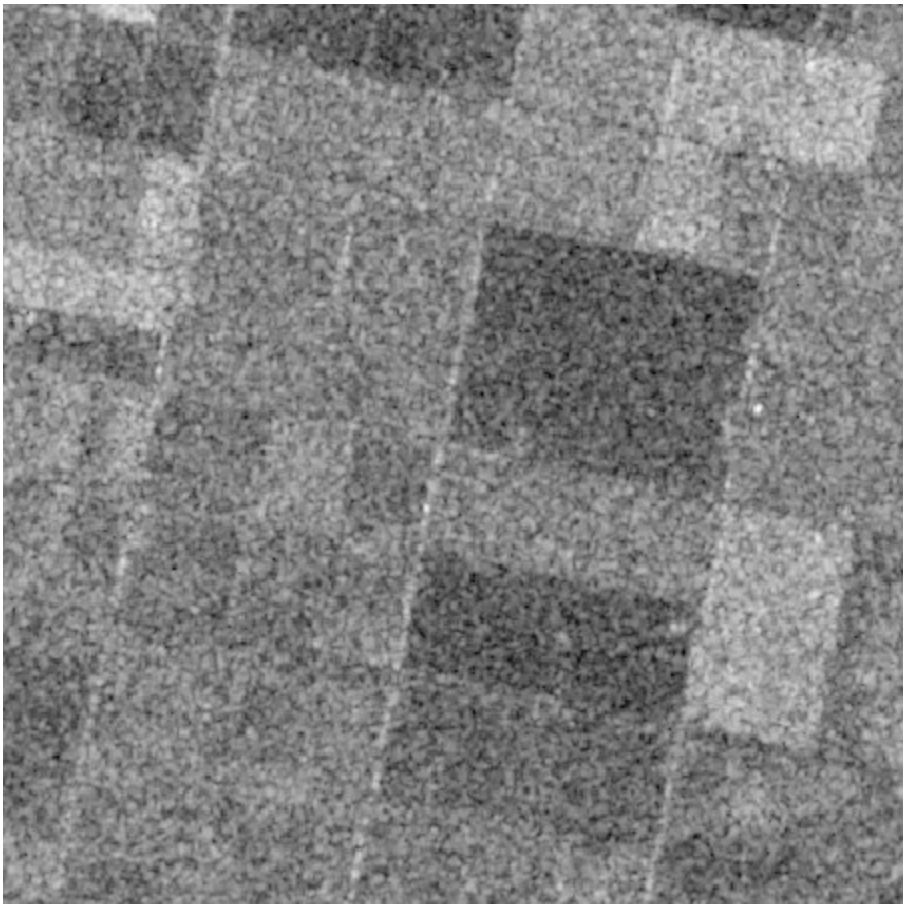


VV (dB), display range -25 to 0

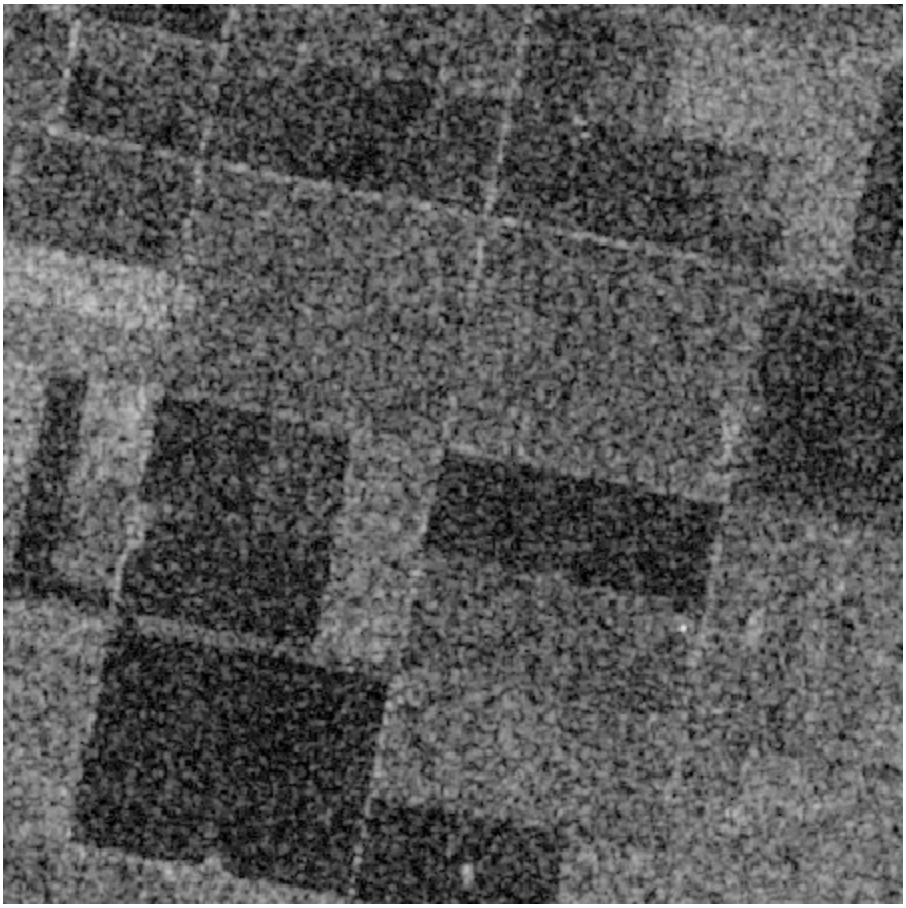


VH (dB), display range -30 to -5

Argentina field 52 (2021-10-25, descending):

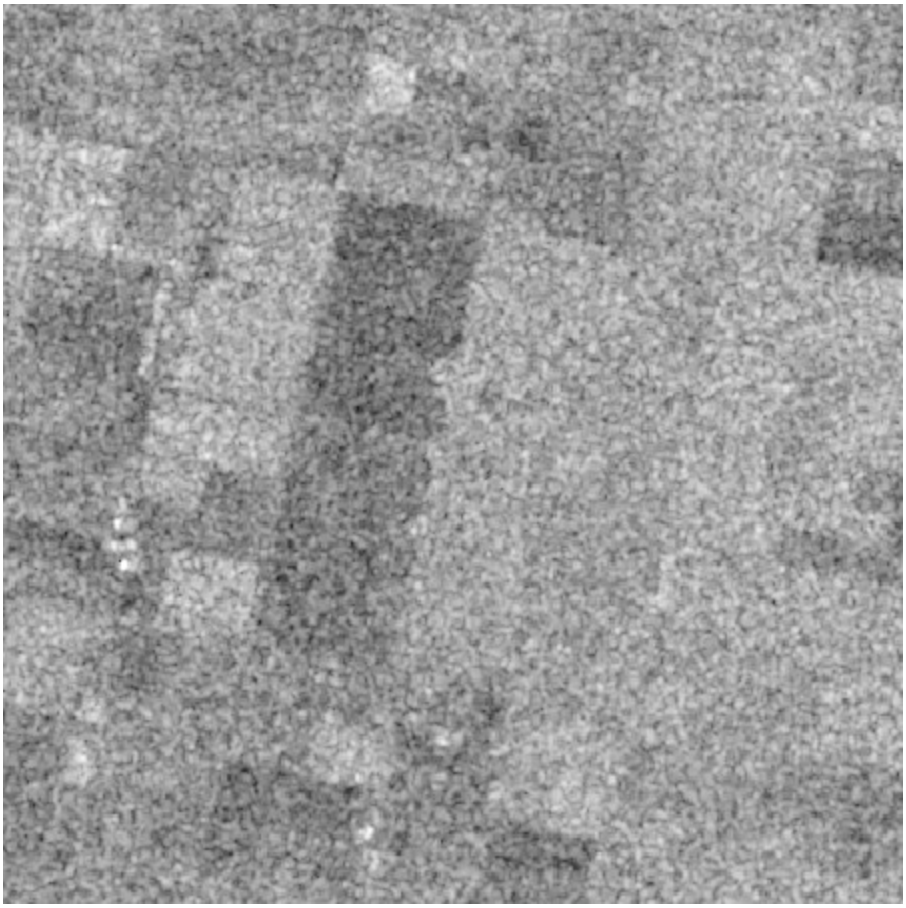


VV (dB), display range -25 to 0

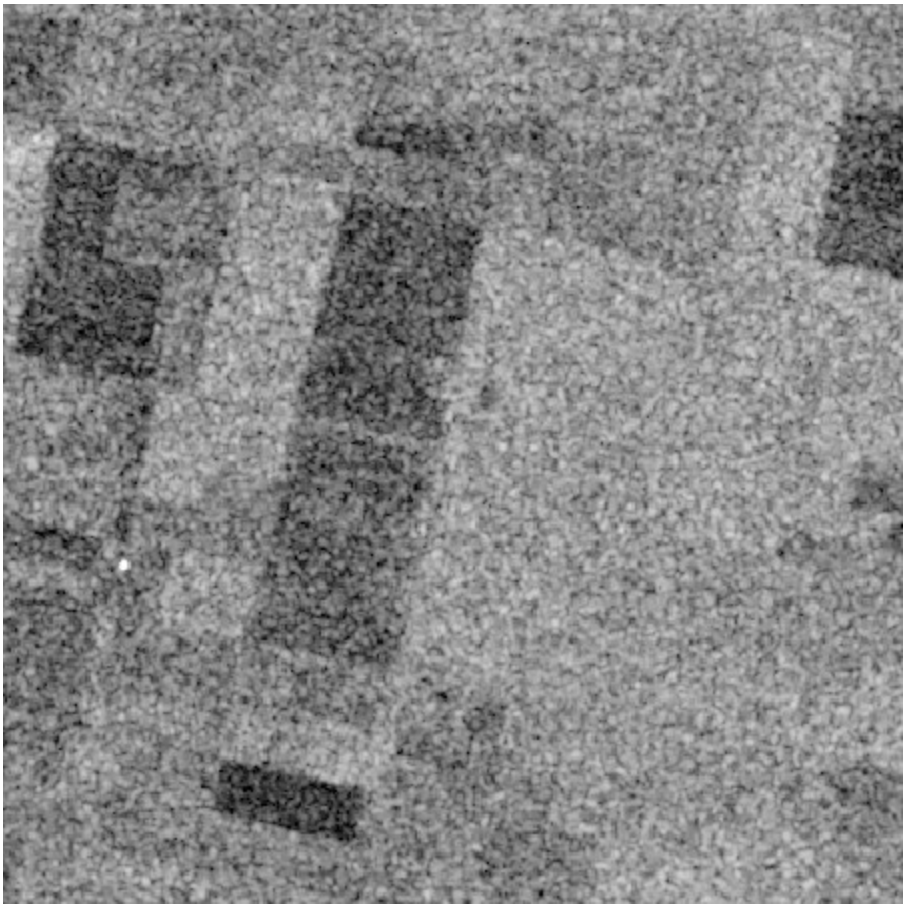


VH (dB), display range -30 to -5

Argentina field 468 (2021-04-16, descending):



VV (dB), display range -25 to 0



VH (dB), display range -30 to -5

Sentinel-1 VV and VH for each of the three fields, the date of the image and the descending orbit and 10 m resolution in each caption. Display ranges: VV from −25 to 0 dB and VH from −30 to −5 dB.]The first attempt to extract did not return any valid values for any record. This was due to the imported tables not having a longitude or latitude column in them and thus Earth Engine added in a blank geometry to all of the records. For 50 samples, it was solved by decoding the coordinate vectors stored. The other 50 samples from Brazil (field 96, 117) and Argentina (field 557) (from 2016 to 2017) are currently pending as the coded location of the sample has multiple possible solutions and could not be resolved using Sentinel-2 within the project timing. They were not assigned any SAR values and no estimates were filled in.Values are all point and not field averages. Coordinate data were assumed to be coded from the information provided by the dataset and the information source does not provide documentation of this, so it is recommended that the provider be contacted to confirm.

6. Learnings:

- We discovered how to programatically review satellite data and ensure completeness in the field identifiers, observation date, and coordinate data.

- We learned how to calculate NDVI using near-infrared bands.

- We also discovered how to use histograms and box plots to see data patterns and compare crop categories. Also SAR data must line up before their features can be mixed.

- Experience was gained in using Python to generate an initial validation report and creating visualisations of the distribution of crop classes.

- Checking the completeness of the metadata was a different question from the actual spatial and temporal periodicity of SAR and optical satellite observations.

- Don't worry about missing geometry, that's not missing data. We were expecting a record to be missing, but none was found in Earth Engine, but there was still a record present in the encoded unit vectors. We learned how to check what columns represent on a table before we deem it unusable.
- There must be independent data to test an inferred transformation. The coordinates were then translated and a number of possible locations were suggested, the only way to discern which was correct was to compare them with the actual Sentinel-2 values. If it wasn't for that check we would have taken SAR from the incorrect location and not realized it.
- It is unlikely that the same sensor will observe on the same day. The Sentinel-1 nearest date used was 2.6 to 5.6 days before or after the dataset date, so both dates are presented in the output and we mentioned the matching rule, so that the SAR and optical features are in the same game later in the output.

7. **References**:

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